

IoT BASED SMART VEHICLE SECURITY AND SAFETY SYSTEM



Report submitted in the partial fulfilment of the requirements for the degree
of B.Sc. Telecommunication Engineering

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DECLARATION

We hereby declare that this project report/thesis entitled “IoT Based Smart Vehicle Security and Safety System” submitted to the “Department of Telecommunication Engineering”, is a record of an original work done by us under the guidance of Supervisor “Dr. Sajjad Ali” and that no part has been plagiarized without citations. Also, this project work is submitted in the partial fulfilment of the requirements for the degree of Bachelor of “Telecommunication Engineering”.

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FYDP REPORT APPROVAL

The Department of Telecommunication Engineering, University of Engineering & Technology Mardan accepts the report in the present form and it satisfies the entire requirements of the Final Year Project (FYP).

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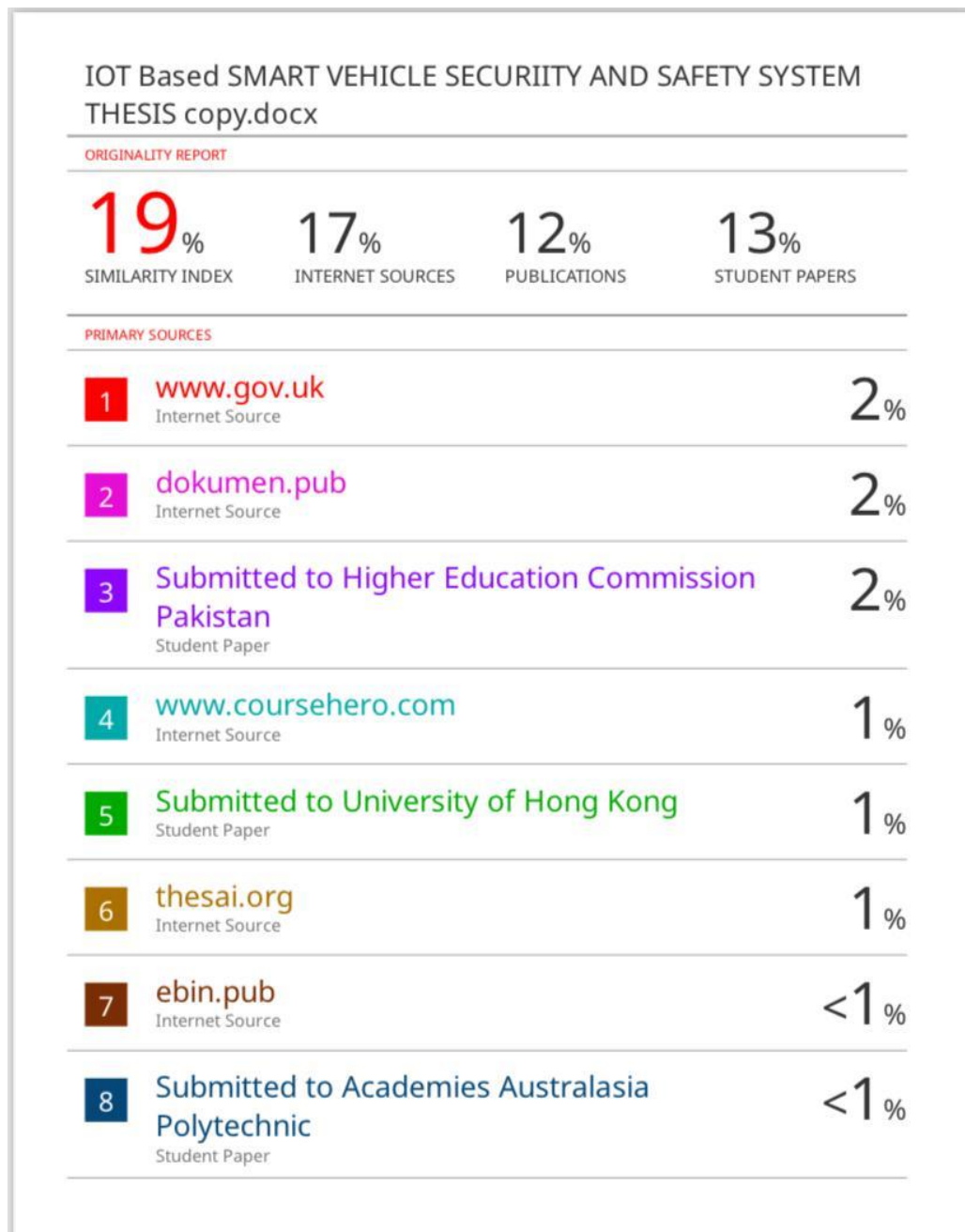
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ABSTRACT

The increasing rise in traffic accidents and vehicles thefts requires the development of modern vehicle security and safety systems. The suggested IoT-based smart vehicle security and safety system seeks to address these challenges by combining several safety and security features into a single, inclusive solution. The goal of this project is to ensure that only alcohol-free and attentive drivers are allowed to operate the vehicle, while also improving passenger safety and post-crash response times.

The Internet of Things (IOT) is making all the things easy for human lives in every aspect. IOT is a network of correlative devices that connects or links through the internet, so they can communicate with each other. IoT refers to a network of physical objects or things that are embedded with sensors, software and connectivity capabilities allowing them to collect or exchange data over the internet. As IoT has many uses that includes smart automobiles and their safety, security, GPS or navigation system, and efficient fuel consumption. As vehicular accidents, thefts, and safety concerns continue to pose significant challenges in the country and this initiative presents an innovation solution that only addresses these issues that also sets a benchmark for the future of automotive technology. As the proposed system utilizes an embedded system based on the internet of things (IoT) and the Global System for Mobile Communication (GSM). For avoiding an accident, when the system is activated, we install a pressure sensor for checking the seat belt, if the person or driver is not wearing a seat belt; the ignition will not be turned on. After that we install the alcohol sensor that detects the alcohol usage and if the sensor detects any alcohol consumption, the ignition or engine will not be turned on. If the driver is feeling sleepy or drowsy, the eye-blink sensor detects the drowsiness of the driver and then the sensor will turn off the engine. And in the end if there are any accident detects, the GPS module will send a location to the contact which is mentioned by using the GSM module.

Plagiarism Check Report



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LIST OF ABBREVIATIONS

Abbreviation	Expansion
IoT	Internet of Things
GSM	Global System for Mobile Communication
GPS	Global Positioning System
ADAS	Advance Driver Assistance System
NHTSA	National Highway Traffic Safety Administration
ECU	Engine Control Unit
CADD	Community Against Drunk Driving
PWM	Pulse Width Modulation

TABLE OF CONTENTS

Contents

Declaration.....	Error! Bookmark not defined.
FYDP Report Approval	Error! Bookmark not defined.
Abstract	Error! Bookmark not defined.
Acknowledgements	Error! Bookmark not defined.
List of Abbreviations.....	Error! Bookmark not defined.
Table of Contents	Error! Bookmark not defined.
List of Figures.....	Error! Bookmark not defined.
List of Tables	Error! Bookmark not defined.
Chapter 1: Introduction	Error! Bookmark not defined.
1.1 Overview	Error! Bookmark not defined.
1.2 Problem Statement	Error! Bookmark not defined.
1.3 Objective	Error! Bookmark not defined.
1.4 The Project Target	Error! Bookmark not defined.
1.5 Project Scope	Error! Bookmark not defined.
Chapter 2: Literature Review	Error! Bookmark not defined.
2.1 Project Description	Error! Bookmark not defined.
2.2 Previous works.....	Error! Bookmark not defined.
Chapter 3: Project Methodology and Implementation	Error! Bookmark not defined.
Bookmark not defined.	
3.1 Project approach (Overall flow diagram).....	Error! Bookmark not defined.
3.2 Diagram.....	Error! Bookmark not defined.
3.3 Design procedure	Error! Bookmark not defined.
3.3.1 Component selection	Error! Bookmark not defined.
3.3.2 Arduino uno	Error! Bookmark not defined.
3.3.2.1 Specifications	Error! Bookmark not defined.
3.3.3 Seat Belt Sensor	Error! Bookmark not defined.
3.3.4 Alcohol Detection Sensor	Error! Bookmark not defined.
3.3.5 Eye Blink Sensor.....	Error! Bookmark not defined.
3.3.6 GSM Module	Error! Bookmark not defined.
3.3.6.1 Specifications of GSM Module	Error! Bookmark not defined.
3.3.7 GPS Module.....	Error! Bookmark not defined.
3.3.7.1 Specifications of GPS Module	Error! Bookmark not defined.
3.4 Hardware setup	Error! Bookmark not defined.
3.4.1 Hardware Circuit Diagram.....	Error! Bookmark not defined.

3.5	Programming	Error! Bookmark not defined.
3.6	Alert mechanism.....	Error! Bookmark not defined.
Chapter 4: Results and Discussion.....		Error! Bookmark not defined.
4.1	Detection of Seat Belt Non-Compliance.....	Error! Bookmark not defined.
4.2	Drunk Driving Avoidance.....	Error! Bookmark not defined.
4.3	Drowsiness Detection and Alert	Error! Bookmark not defined.
4.4	Collision Detection and Emergency Notification.....	Error! Bookmark not defined.
4.5	Alcohol/drugs, speed and fatigue in fatal crashes (2020 - 2022)	Error! Bookmark not defined.
4.6	Deaths in crashes involving driver fatigue since 1983.....	Error! Bookmark not defined.
4.7	Causes of Crashes.....	Error! Bookmark not defined.
4.8	Discussion	Error! Bookmark not defined.
4.8.1	Integration of Results with Project Objectives...	Error! Bookmark not defined.
4.8.2	Limitations and challenges	Error! Bookmark not defined.
Chapter 5: Conclusions and future work		Error! Bookmark not defined.
5.1	Conclusion	Error! Bookmark not defined.
5.2	Summary of results.....	Error! Bookmark not defined.
5.2.1	Seat Belt Usage Compliance.....	Error! Bookmark not defined.
5.2.2	Drunk Driving Prevention	Error! Bookmark not defined.
5.2.3	Drowsiness Detection	Error! Bookmark not defined.
5.2.4	Emergency Response.....	Error! Bookmark not defined.
5.3	Future Work and Research.....	Error! Bookmark not defined.
5.3.1	Addressing Mortality in Road Accidents.....	Error! Bookmark not defined.
5.3.2	Sensor Optimization	Error! Bookmark not defined.
5.3.3	Data Analytics and Predictive Modelling.....	Error! Bookmark not defined.
5.3.4	Integration with Advanced Vehicle Technologies	Error! Bookmark not defined.
References		Error! Bookmark not defined.

LIST OF FIGURES

Figure 3-1 Overall flow diagram of IoT based Smarty Vehicle Security and Safety System	1
Figure 2 Activity Diagram.....	1
Figure 5 1	
Figure 6 1	
Figure 7 GPS Module.....	1
Figure 8 1	
Figure 9 1	
Figure 10	1
Figure 11	1

LIST OF TABLES

Table 1 Specifications of Arduino Uno	Error! Bookmark not defined.
Table 2 GSM Module Specifications	Error! Bookmark not defined.
Table 3 GPS Module Specifications	Error! Bookmark not defined.
Table 4	Error! Bookmark not defined.
Table 5	Error! Bookmark not defined.

Chapter 1: Introduction

1.1 Overview

The Internet of Things (IoT) is making all the things easy for human lives in all aspects. The (IoT) is a revolutionized technical development that allows physical things to be connected through internet for data collection and sharing. [1] This network of interconnected things includes transportation, industrial machinery, and smart household items. Nowadays, everyone used the internet to communicate and exchange information. For example, IoT enables automated control of security, heating, and lighting systems in smart houses, enhancing comfort and reducing energy consumption in the automotive sector. Car performance and safety are enhanced via the Internet of Things. In order to provide patients with more effective care, wearable technology is also being used in healthcare procedures and data transmission to healthcare providers. The Internet of Things has made life easier, more efficient, and more convenient overall as these devices are interconnected. [2]

Internet of Things (IoT) solutions are now widely used in various industries during this rapid period of technical advancement and innovation. This denotes the start of a new phase characterized by enhanced connectivity, insights derived from data, and automated procedures. Smart homes and city infrastructure, as well as healthcare and industrial processes, have all benefited from IoT technologies, which have altered our interactions with the outside world and increased productivity, sustainability, and efficiency. [3]

Nowadays the risk to people is quite significant. It is because of the lack emergency facilities available in our country. The state of affairs in Pakistan at the moment indicates a serious risk to human-life, especially given the dearth of emergency facilities. Modern civilization depends heavily on transportation, yet it also has inherent vulnerabilities that put lives in risk. There are approximately 29,000 peoples in Pakistan lose their lives in traffic accidents every year serves as a vivid reminder of this predicament. [4] This number is among the highest in the world for road accidents facilities according for 1.93% of all deaths in the nation. It is careless driving that is the main reason of this high rate accident. [5]

Numerous risky action that raises the possibility of collisions and fatalities are included in the category of reckless driving. It is not uncommon to witness drivers in Pakistan committing dangerous manoeuvre, such speeding excessively, which shortens their reaction time in the event of an emergency.

Road traffic accidents are the main cause of death for young people and the eighth major cause of all deaths globally with predestined 1.24 million deaths every year. About 85% of deaths are occurring in developing countries. Males, especially those between 15 and 44 years old, are the extremely affected group of people with traffic accidents. [6] Overtaking too closely is another problem. Drivers frequently try to pass other car without paying attention to the other traffic or the state of the road, which can result in serious accidents such a head-on collision. During under the influence of alcohol or drugs makes the issue worse. [7] A driver with substance impairment struggle to handle their car and reaction time, distorts their judgment and weakens their motor abilities. During the influence of drinking makes the issue worse. The issue is further exacerbated by the absence of emergency response facilities. Victims in accidents or crashes frequently do not obtain prompt medical care, which might mean the difference between life and death. Furthermore, Pakistan poor roads infrastructure frequently plays a role in the country's high accident rate. Extreme weather conditions increase the poor state of neglected roads, which are especially prevalent in rural regions.

The fatigue is a main contributor to road safety. Due to fatigue the police reported 100,000 crashes per year are directly caused by fatigue, which is significant figure. according to mindful estimate from by the American National Highway Traffic Safety Administration (NHTSA). These accidents result in more than 1,550 fatalities and 71,000 injuries. [8] The real numbers may be much higher, however, as there is no way to test whether a driver is drowsy. Fatigue impairs decisions- making abilities, attention spans and response times- all of which are critical for safe driving. Despite these concerns, a lot of drivers tend to overestimate their capacity to stay awake and aware or underestimate the risks with driving when fatigue. Driver fatigue is caused by a number of things. Common reasons include driving at night, sleeping irregularly and long stretches of time spent behind the wheel without any enough rest. In addition, using drugs

that make you drowsy and certain medical disorders like sleep apnoea raise the risk. Life style elements, such as getting to little sleep because of obligations at work or home, can also play a significant role.

The Global system for mobile communication (GSM) and the Internet of Things (IoT) are two examples of cutting- edge technology that the proposed system integrates to dramatically improve vehicle security and safety. The vehicle network of interconnected gadgets, which includes a variety of sensors including the accelerometer and GPS module are used by this system. With real-time communication to a central server or cloud platform, these sensors continually gather data several factors, including position, speed and environment conditions. [9] Abnormalities are found by analyzing the data, such as unexpected vehicle movements, abrupt deceleration suggestive of crashes or abnormalities that could indicate security or safety issues. The (IoT) make it possible for these gadgets to communicate with each other seamlessly and guarantee continuous vehicle conditions monitoring. Because of this connectedness, problems can be identified right away and alarms maybe send out quickly.

Furthermore, the GSM capability of the system guarantees dependable connection between the car and outside monitoring system. This includes the capability for car owners to use web interface or smartphone app to remotely access and manage their vehicles. [10] Owners may control their vehicle conditions in real time, including its position, fuel levels and whether the alarm has gone off. They can also lock or unlock their doors and start or stop the engine. Additionally, the system has an emergency call (eCall) capability that lets the driver manually place an emergency call in case it's necessary. This adds an extra degree of security, guaranteeing that assistance may be called in the event of an emergency of any kind, even if the automated system malfunction.

The suggested system includes an alcohol detecting device and seat belts to improve vehicle safety and lower the likelihood of accidents. [11] A pressure sensor built into the seat determines if the driver is using a seat belt when the systems is turned on. [12] Before the car's ignition system is turned on, this sensor makes sure the seat belts are securely tightened. This technology ensures that the driver cannot start the vehicle if the pressure sensor detects that the seat

belt is not properly tightened. To protect the driver and passengers, this feature not only encourages safe driving practices, but also ensures that local regulations in many locations are followed before the vehicle is operational.

The given system has an alcohol sensor in addition to the seat belt check to detect drunk driving. The alcohol sensor detects any indication of alcohol intake in the driver's breath after the seat belt has been verified to be securely secured. The ignition system of the car is instantly turned off by the sensor if it finds any alcohol present at all, which keeps the engine from starting. [13] This two-pronged safety strategy, which mandates the use of seat belt in addition to a positive alcohol test, greatly lowers the likelihood of accidents brought on by drunk driving. By incorporating these sensors, the guarantees that the car can only be driven by a sober and seat- belted driver, improving general road safety and averting possible injury to the driver and other road user.

The eye- blink sensor uses advanced algorithms to examine blink patterns, length and frequency when it notices indication of fatigue or drowsiness in the driver. This study assists in differentiating blink that are symptomatic of exhaustion from those that are normal. The sensor can detect minutes alternation in the driver behaviour that can indicate a reduction in attentiveness by continually tracking the driver eye movement. This sensor starts a sequence of steps to maintain road safety as soon as it detects a significant decline in the driver's level of attentiveness. [14] First, the sensor initiates a controlled shutdown procedure by interacting with the engine control unit (ECU) of the car's onboard computer system. In this method, the engine fuel supply is turned off as the engine's power output is steadily reduced. The system's goal is to minimize disruption and guarantee a safe transition to a complete stop by progressively slowing down the vehicle instead of halting suddenly. The system serves as a protective defence against the risks sleepy driving by combining these cutting- edge technology and safety procedures, greatly lowering the likelihood of accidents brought on by driver weariness.

As the vibration sensor includes into the car's safety system serves as an essential early warning system in the case of obstruction or possible crashes. As soon as it notices abrupt movement or impact that may be sign of an accident, it starts a

series of procedures to reduce the danger and alert the appropriate parties. [15] In order to notify pre-designated emergency contact of the impediments or collision in the real time, the sensor first sends a message to them. This message alert anybody the driver has specified, such family members or emergency contact, about the possible hazard. The driver is simultaneously alerted to the identified hazards and is prompted to take timely action if feasible by means of the buzzer system of the car. Through the utilization of GPS technology, the GSM module reliably transmits the exact location of the car at the moment of impact, enabling emergency personnel act quickly and precisely. [16] In the addition to improve the overall vehicle safety, this seamless integration of vibration sensors and GSM technology allows for quick emergency reactions in dire circumstances, potentially saving lives and minimizing the impact of accidents.

1.2 Problem Statement

Road safety is a serious concern around the world, with vehicle accidents, thefts, and driver-related safety issues all contributing to injuries and fatalities. Despite advances in automotive technology, numerous critical safety issues remain unaddressed. These challenges include:

- **Seat Belt Non-Compliance:** A significant number of car occupants neglect to use seat belts, which are an important safety precaution to prevent against injuries in an accident. Traditional systems lack enforcement tools to guarantee that seat belts are used before driving the car.
- **Drunk Driving:** Drunk driving is a significant cause of road accidents. Current preventive measures, such as random Breathalyzer tests, are insufficient to reduce the likelihood of drunk driving accidents. There is a need for a real-time, in-car technology that can detect alcohol levels in drivers and stop the vehicle if intoxication is detected.
- **Drowsy Driving:** Driver tiredness is another key factor in traffic accidents. Existing technologies, such as lane changing alerts, do not sufficiently address the underlying cause: driver fatigue. An integrated

solution that monitors driver attention and steps in when necessary is important.

- Accident Notification and Emergency Response: Immediate and accurate notification of accident information to emergency services is critical for providing prompt medical care. Many automobiles lack automatic collision detection systems that communicate important information to emergency responders, resulting in delayed rescue operations and greater death tolls.

Addressing these difficulties demands an innovative solution that makes use of cutting-edge Internet of Things (IoT) and mobile communication technology. This project proposes creating a comprehensive IoT-based Smart Vehicle Security and Safety System that will enforce seat belt compliance, prevent drunk and drowsy driving, and give emergency services with real-time accident notification and location information. This system attempts to improve vehicle safety by combining numerous sensors and communication modules, lower the chance of human-caused accidents, and provide timely emergency response in the case of an accident.

1.3 Objective

Our aim in this project is to achieve the following objectives:

- The engine or ignition will only be turned on if the seat belts are locked in.
- As we install a gas sensor in the system that detects either the driver is drunk or not. If the driver is not drunk, only then will be the engine ignites and turned on.
- To check the drowsiness of the driver, we install an eye-blink sensor that will ensure that the driver is not drowsy, and then the engine will be turned on.
- The switches are used to detect the accident. To assure post-crash safety, an alert system is used which uses a GPS system to determine the geographical location of the crashed car and sends the notification to the provided contact.

1.4 The Project Target

The major target of this initiative is to improve vehicle safety by addressing crucial issues such as seat belt use, drunk driving, drowsy driving, and effective post-accident response. This technology is intended for vehicle manufacturers hoping to include modern safety features into new vehicles, giving them a competitive advantage in the market. It also targets car owners and drivers with a solution that improves personal and passenger safety while providing peace of mind through automated safety inspections and emergency notifications.

Furthermore, the project helps regulatory organizations and traffic authorities with their road safety activities by delivering technology that enforces safety rules and decreases traffic accidents. The system's goal is to provide useful data and insights that will help shape future road safety policies and tactics. Overall, the project aims to have a significant impact on vehicle safety, reducing accidents and saving lives.

1.5 Project Scope

The IoT-Based Smart Vehicle Security and Safety System intends to integrate a variety of advanced technology in order to dramatically improve vehicle safety.

The project requires the creation of various critical components. First, it comprises the development of a seat belt compliance system that assures the vehicle's engine will only start if the seat belts are securely attached. Sensors for detecting seat belt usage and an ignition interlock device are integrated. Next, the project will include an alcohol detecting system. This device will employ sensors to detect alcohol in the driver's breath. If alcohol has been found, the vehicle's ignition will be stopped preventing the driver from using it while intoxicated. Another important feature is the sleepiness detecting mechanism. The system will use eye blink sensors to monitor the driver's attention. If the system detects drowsiness, it will inform the driver and may disable the car to prevent fatigue-related accidents. Additionally, the project will provide an accident detection and warning system. This system will detect collisions with switches and locate the car using GPS modules. In the case of an accident, the system will immediately transmit messages to emergency services and designated contacts via GSM technology.

The project also includes the creation of a central processing unit that will combine and evaluate data from all sensors. It will implement strong communication protocols like Wi-Fi, cellular, and Bluetooth to enable real-time data transmission and alarms. In addition, a user interface will be developed to enable system monitoring and control, as well as feedback mechanisms to continuously enhance system performance based on user experience and data analysis.

Chapter 2: Literature Review

2.1 Project Description

The IoT-Based Smart Vehicle Security and Safety System is an innovative initiative that combines modern Internet of Things (IoT) technologies to improve vehicle safety and security. This system addresses several major difficulties in modern road safety, including as seat belt compliance, driving while intoxicated, fatigued driving, and a lack of immediate accident reporting. The project includes the creation and installation of a variety of sensors and communication modules to form a comprehensive safety system. The system contains a seat belt sensor, which prevents the vehicle's engine from starting unless the seat belts are correctly tightened. An alcohol detection sensor checks the driver's breath for alcohol and disables the ignition if it is found. An eye-blink sensor detects indicators of driver fatigue and alerts the driver or disables the vehicle if necessary. In addition, the device includes a switches to detect collisions. In the case of an accident, the system uses GPS technology to pinpoint the vehicle's location and sends messages to emergency personnel and specified contacts via GSM. This ensures a quick response and increases the possibility of prompt help.

The system's central processing unit integrates and analyzes data from all sensors, providing reliable connection protocols like Wi-Fi, cellular, and Bluetooth for real-time data transfer and warnings. A user interface is created to enable system monitoring and control, as well as feedback mechanisms to continually enhance system performance based on user experience and data analysis.

Overall, the IoT-Based Smart Vehicle Security and Safety System seeks to dramatically reduce the risks associated with traffic accidents, improve post-accident response times, and improve overall vehicular safety, thereby contributing to safer roads and saving lives.

2.2 Previous works

As the study presents a statistical analysis of highway fatalities in 2002, emphasizing the essential importance of seat belt use in avoiding deaths. In 2002, traffic accidents killed 43,005 people. Of these victims, 32,598 had comprehensive information on seat belt usage. Among them, 19,103 people were not wearing seat belts when the disaster occurred. The report emphasizes the potential life-saving impact of increasing seat belt use, estimating that if 90% of Americans consistently used seat belts, up to 42,000 lives may be saved each year. This statistic highlights the significance of seat belt promotion and enforcement as a critical public safety measure. [17]

In [18] the author addresses the serious consequences of car accidents in India, citing an annual death toll of 134,000. According to the Community Against Drunk Driving (CADD), alcohol is a key factor, accounting for 70% of these accidents. In smaller cities, the impact of alcohol on car accidents ranges from 44% to 67%. Since 2001, the number of drunk driving accidents has increased drastically, seven times in Delhi and more than sixteen times in Mumbai. This research demonstrates a vital need for stricter restrictions against drunk driving to improve road safety in India.

In another study the author discusses an innovative technology for preventing alcohol-related auto accidents. This system includes an alcohol detector, which informs the driver when excessive alcohol is detected. The information is presented on an LED panel, and a buzzer sounds a warning, reminding the driver to reassess their situation before making any vital judgments. The system's CPU is an Arduino, which has certain performance limitations. This device is intended to improve road safety by providing real-time feedback to potentially intoxicated drivers. [19]

Another study describes how a MQ-4 gas module, which detects gas levels, is installed on a PCB board with a 5V DC working voltage. The sensor output values can be acquired in both analog and digital formats. This configuration allows for flexible and accurate monitoring of gas concentrations, making it suited for a variety of applications that require gas detection and measurement. [20]

Moreover, another research describes a multi-sensor system meant to improve driver safety by monitoring various crucial elements. An eye blink sensor detects driver fatigue, a proximity sensor detects surrounding obstacles in front of the mirror, and an alcohol sensor examines whether the driver is under the influence of alcohol or not. This comprehensive approach seeks to prevent accidents by addressing the various possible risks connected with intoxicated driving. [21]

In [22] the author says, The IR sensor, the relay and the eye blink sensor frame make up this module. The eye blink sensor detects when the driver's eyelids are closed by sending and receiving infrared rays. If the driver falls asleep or an accident occurs, a vibrator device is activated to notify the driver. The IR receiver collected the reflected rays when the driver eyes are closed and IR transmitter send the IR rays in the direction of driver eyes. The relay supplies the additional current required for the module, which is coupled to a microcontroller board to manage the system's tasks.

Another study investigates an innovative strategy to improving road safety by implementing an enhanced driver monitoring system. The primary component of this system is an eye blink sensor that detects fatigue or sleepiness in the driver. When the sensor detects indicators of drowsiness, it activates an automatic engine shut-off mechanism, avoiding any accidents caused by drowsy driving. This article examines the creation, installation, and effectiveness of this revolutionary technology, with the goal of contributing to ongoing efforts to improve road safety and reduce traffic fatalities. [23]

In [24] the author describes an innovative approach to addressing the essential issue of road safety using embedded systems driven by IoT (Internet of Things) and GSM (Global System for Mobile Communications) technology. The suggested technology combines automotive proximity sensors within vehicles to detect obstructions in their path, with the goal of reducing accidents and increasing driver awareness. This revolutionary system uses IoT and GSM technologies to provide real-time monitoring and notifications, allowing for proactive interventions to reduce collision risk and improve overall road safety. This study investigates the creation, implementation, and efficacy of this cutting-edge approach, which contributes to improvements in accident prevention and driver safety on the road.

There is another study proposes a comprehensive automotive safety measure that includes a GPS-based alert system to provide continuous monitoring of the vehicle's location, particularly in post-accident scenarios. This technology provides an extra layer of safety by tracking the car's location even in the case of an accident. This system uses GPS technology to offer real-time position data, allowing for fast emergency response and protecting the safety of passengers and occupants. [25]

Another study describes the core principles of the Global Positioning System (GPS), which uses exact timing and known positions of specialized satellites to correctly calculate location. GPS satellites continuously broadcast their positions and time signals, allowing GPS receivers to determine their precise location and temporal precision. To accomplish this, GPS receivers track signals from numerous satellites and solve mathematical equations to calculate their position and departure from real-time. Notably, the receiver must receive signals from at least four satellites in order to accurately compute four unknown values. This study goes into the fundamental principles and operating mechanisms of GPS technology, emphasizing its importance in a variety of applications, including navigation, surveying, and beyond. [26]

In [27] the author describes a revolutionary approach to vehicle safety systems, emphasizing the need of real-time accident detection and communication. It suggests incorporating GSM (Global System for Mobile Communications) technology with automotive systems to quickly notify necessary parties in the case of an accident and send the vehicle's geographical coordinates. Drawing on existing literature, the paper investigates the use of similar approaches to measure cow health indices. By using these insights, the article hopes to provide a novel way for improving vehicle safety and comprehending key concepts in adjacent domains. This introduction lays the groundwork for a thorough examination of the proposed system's development, implementation, and potential impacts.

Chapter 3: Project Methodology and Implementation

The IoT-Based Smart Vehicle Security and Safety System project aims to provide a comprehensive and dependable safety system that solves crucial issues such as seat belt compliance, alcohol consumption, driver fatigue, and accident detection. Using IoT technology, the initiative intends to greatly improve car safety and security, going beyond typical automotive safety measures.

This project focuses on the careful selection of hardware components, such as numerous sensors and communication modules, to ensure effective real-time monitoring and reaction. The system is designed systematically by constructing an integrated architecture that connects sensors to a central processing unit, which then conveys crucial information and alarms via existing communication protocols such as Wi-Fi, cellular networks, and Bluetooth. The development process includes comprehensive prototypes, in which each component's functioning is tested individually and as a whole. Intensive testing phases are carried out to ensure that the system functions consistently under a variety of circumstances and scenarios. Data analysis is also an important part in this project since the system continuously receives and processes data from the sensors to monitor driver behaviour and vehicle status in real time. This data is utilized to provide useful information and alerts, help in the prevention of accidents and ensure prompt response when necessary.

Finally, this technology gives real-time information on numerous safety parameters, allowing drivers and emergency personnel to make informed judgments promptly. This preventive strategy improves not only individual car safety, but also general road safety by lowering accident rates and improving emergency response times. This initiative intends to set a new standard for vehicles security and accident avoidance by using IoT technologies to upgrade vehicle safety.

3.1 Project approach (Overall flow diagram)

The IoT-Based Smart Vehicle Security and Safety System's operational framework can be seen in the project's methodology's overall flow chart. The following illustration shows how the system's data collecting, processing, and analysis steps are carried out in a systematic manner. It offers a comprehensive overview of the ways in which different parts like GPS modules, GSM communication modules, drowsiness detection sensors, alcohol and seat belt sensors are interconnected to improve vehicle safety. The flow chart shows the systematic procedure by which information is obtained from sensors, sent over communication modules, and instantly processed to guarantee prompt action in the event of a safety hazard. It emphasizes the effectiveness of the system in avoiding collisions and ensuring driver safety by displaying these connections.

This visual assistance shows how IoT technology may be easily incorporated into vehicle systems, improving resource allocation and decision-making for emergency services and drivers both. It clearly shows the project's goal of updating existing vehicle safety techniques, which will ultimately lead to safer roads and a decrease in the effects of accidents.

3.2 Diagram

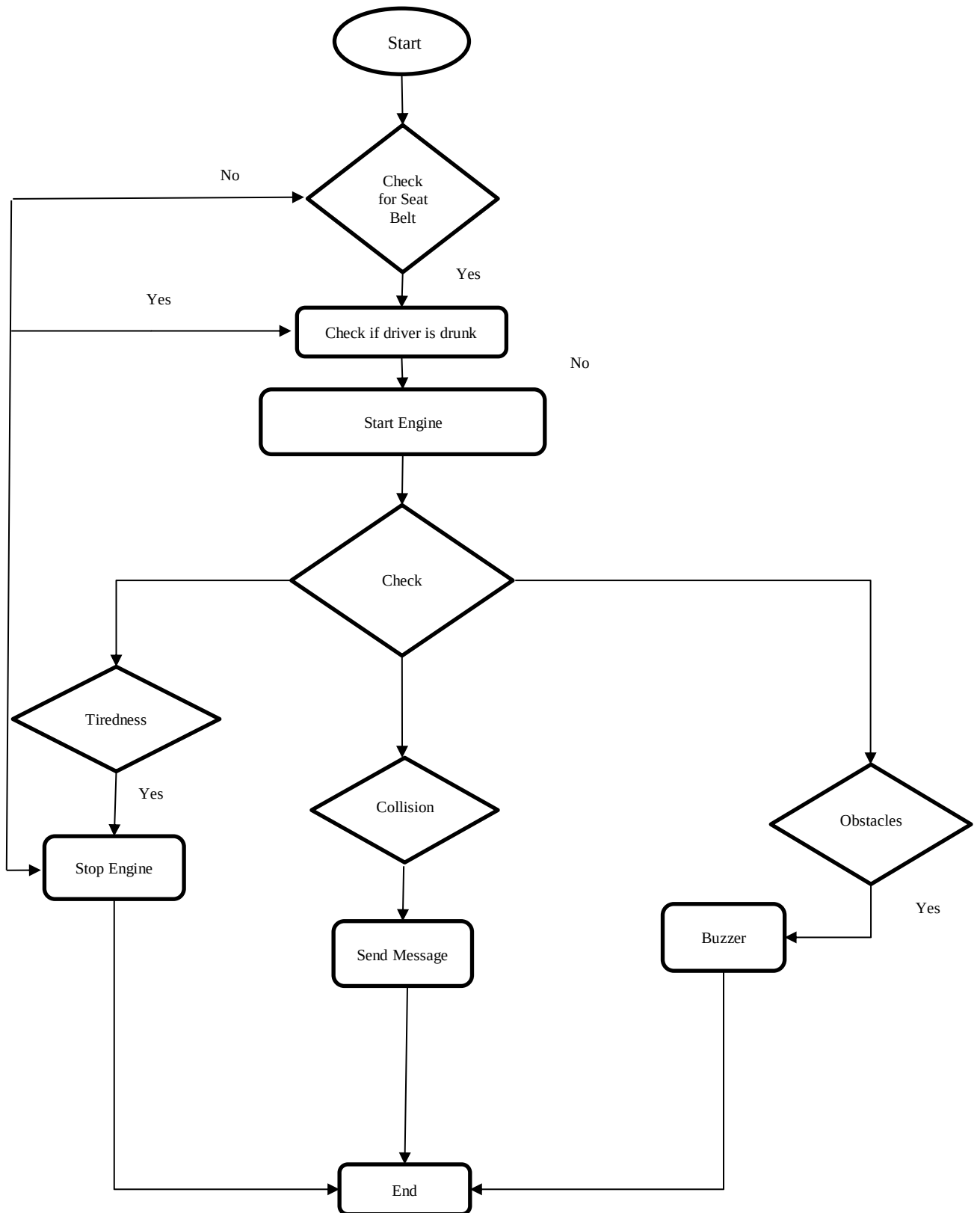


Figure 1 Activity Diagram

3.3 Design procedure

The design process of our IoT-based vehicle safety and security system encompasses several crucial steps, each aimed at ensuring the effectiveness, reliability, and integration of the system. Below, we outline the key aspects of the design procedure:

3.3.1 Component selection

Choosing suitable components is essential to our vehicle security and safety system functioning properly. In order to choose the best sensors, microcontrollers, communication modules, and other hardware components in the market, this step involves comprehensive research and analysis. Performance, reliability, compatibility, and cost-effectiveness are just a few of the factors that are carefully taken consideration during the selection process. Furthermore, simulation tools can be used to evaluate how components behave in various scenarios, ensuring optimal performance and compatibility.

Following comprehensive research and evaluation, the following elements have been selected for implementation into our vehicle's security and safety system:

3.3.2 Arduino UNO

Employing the Arduino Uno microcontroller board, which is well-known for its flexibility in electronics projects, the Internet of Things-based smart vehicle security and safety system is highly adaptable. This system, which is controlled by an ATmega328P microcontroller, achieves an optimal combination of usability and functionality. It is perfect for beginners as well as professionals because of its variety of digital and analog input/output pins, which allow users to easily interface with actuators, sensors, and other electrical components.

Six of the Arduino Uno's 14 digital input/output pins support pulse width modulation (PWM) signals, enabling precise control over motors, LEDs, and various devices. These pins can be set up as inputs to collect data from switches and sensors or as outputs to drive components. Six analog input pins also make it easier to measure voltage from analog sources, which improves the system's capacity to detect and react to external inputs. In addition, the Arduino Uno provides power and ground connections for connecting components and

provides pins specifically for programming and serial communication. The adaptable connectivity options offered by this extensive pin arrangement are crucial for allowing the different IoT-based functions into the smart vehicle security and safety system.

Overall, the Arduino Uno serves as the cornerstone of the IoT-based smart vehicle security and safety system, empowering users to create robust and adaptable solutions to enhance vehicle security and ensure driver safety.

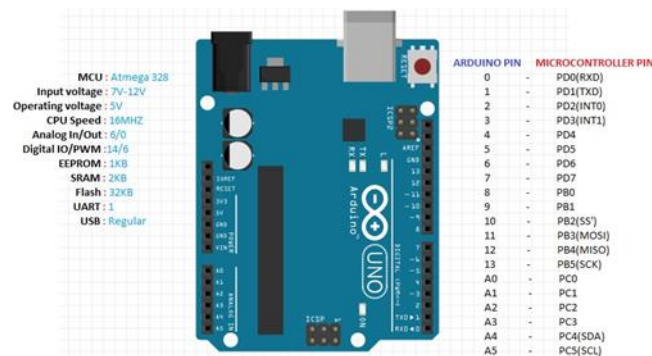


Figure 2 Arduino Uno

3.3.2.1 Specifications

Table 1 Specifications of Arduino Uno

Microcontroller	ATmega328
Clock Speed	16MHz
Operating Voltage	5V
Maximum supply Voltage (not recommended)	20V
Supply Voltage (recommended)	7-12V
Analog Input Pins	6
Digital Input/Output Pins	14
DC Current per Input/Output Pin	40mA
DC Current in 3.3V Pin	50mA
SRAM	2 KB
EEPROM	1 KB
Flash Memory	32KB of which 0.5KB used by boot loader

3.3.3 Seat Belt Sensor

Our project's seat belt sensor makes use of a force-sensitive resistor (FSR), which is a type of pressure sensor. FSRs precisely detect seat belt engagement by measuring changes in resistivity in response to pressure. The sensor is an essential part for improving vehicle safety as it installs into the seat belt buckle or is integrated into the seat belt. It sends out a signal showing the state of the seat belt. Because of its sensitivity, precision, and electrical system compatibility, it is a useful part of our project's safety system.

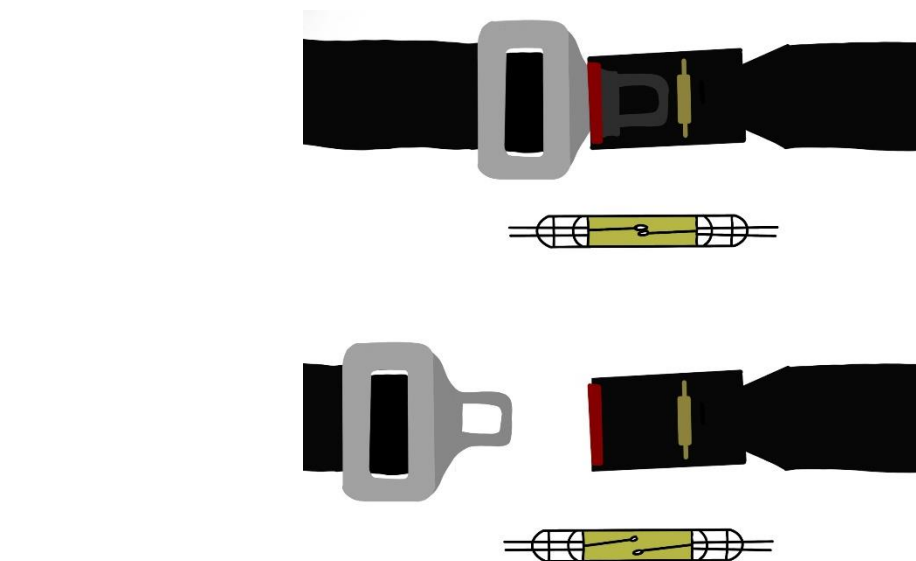


Figure 3
Seat Belt Sensor

3.3.4 Alcohol Detection Sensor

The alcohol detection sensor is essential to improving vehicle safety in our project. This sensor adds an extra degree of security against events involving intoxicated drivers by identifying the presence of alcohol in the driver's breath.

Usually, infrared spectrum or semiconductor-based gas sensors are used by alcohol detection sensors to analyze breath samples and measure alcohol concentrations. The sensor detects alcohol and sets off automatic functions, such turning off the engine or warning the driver when they are driving while intoxicated. For our initiative to be successful in preventing accidents and encouraging safe driving, the alcohol detection sensor's dependability, precision, and responsiveness are critical factors.

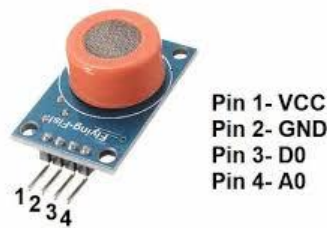


Figure 4
Alcohol Detection Sensor

3.3.5 Eye Blink Sensor

An essential part of our project that promotes driver safety and wellbeing is the eye-blink sensor. This sensor uses patterns in the blinking of the driver's eyes to track their level of awareness. It tracks the frequency and length of eye blinks using optical or infrared sensors to give a real-time assessment of how tired or sleepy the driver is. The sensor can avoid accidents caused by drowsy driving by continually monitoring the motions of the driver's eyes and initiating alarms or intervention actions in response. The eye-blink sensor's dependability and precision are crucial to our project's success in raising traffic safety and averting collisions.



Figure 2
Eye Blink Sensor

3.3.6 GSM Module

A key element of our project is the GSM (Global System for Mobile Communications) module, which permits smooth connection between the vehicle's system and external networks. This module establishes cellular connectivity using GSM technology, enabling the car to send and receive orders, notifications, and data. Our project can communicate critical information in real-time, including location information, emergency alerts, and system status updates, by utilizing GSM networks. The dependability, coverage, and compatibility of the GSM module with current mobile networks are critical to ensure that our technology will effectively improve vehicle safety and communication capabilities.



Figure 3
GSM Module

3.3.6.1 Specifications of GSM Module

Table 2 GSM Module Specifications

Table 1. Specifications of GSM module

Features	Specifications
Operating frequency	GSM 850MHz, EGSM 900MHz, DCS 1800MHz and PCS 1900MHz
Operating Voltage rating	3.2V – 4.8V dc
Output pin voltage	5V dc
Output pin current	25mA
Communication mode	UART interface, configured for full-duplex asynchronous mode
Baud rate	Supports auto bauding, 9.6kb/s used.

3.3.7 GPS Module

The GPS (Global Positioning System) module, which gives the car precise real-time location tracking, is essential. In order to determine the exact geographic coordinates of the vehicle, this module receives signals from GPS satellites. Our system can provide essential features like navigation, route tracking, and emergency location reporting by integrating the GPS module. The GPS module's exceptional precision, dependability, and instantaneous performance are critical to our project's successful enhancement of vehicle safety and fast delivery of location-based information.

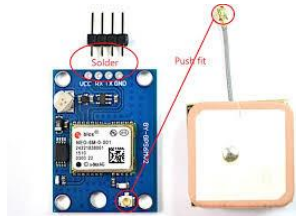


Figure 4 GPS Module

3.3.7.1 Specifications of GPS Module

Table 3 GPS Module Specifications

Hardware	NEO 6M GPS Module
Update Rate	1 – 5 Hz
Sensitivity	-161dBm
UART Baud rate	9600
Operating Voltage	3 – 5 V
Function	Location Tracking

3.4 Hardware setup

The hardware setup for our IoT-based smart vehicle safety and security system started with the assembling of the chosen components into an integrated unit. Every sensor and module were thoroughly attached to the central microcontroller board and incorporated into the vehicle's structure. Carefully thought out wiring and layout were used to guarantee flawless performance in actual driving situations.

Circuit design software was initially used to design the circuit in order to reduce errors and speed up the assembly process. Before actual construction, we were able to optimize the layout, simulate configurations, and confirm component compatibility during this early design process. In doing so, we made sure that the hardware configuration was more dependable and efficient, prepared for successful implementation in the automotive sector.

3.4.1 Hardware Circuit Diagram

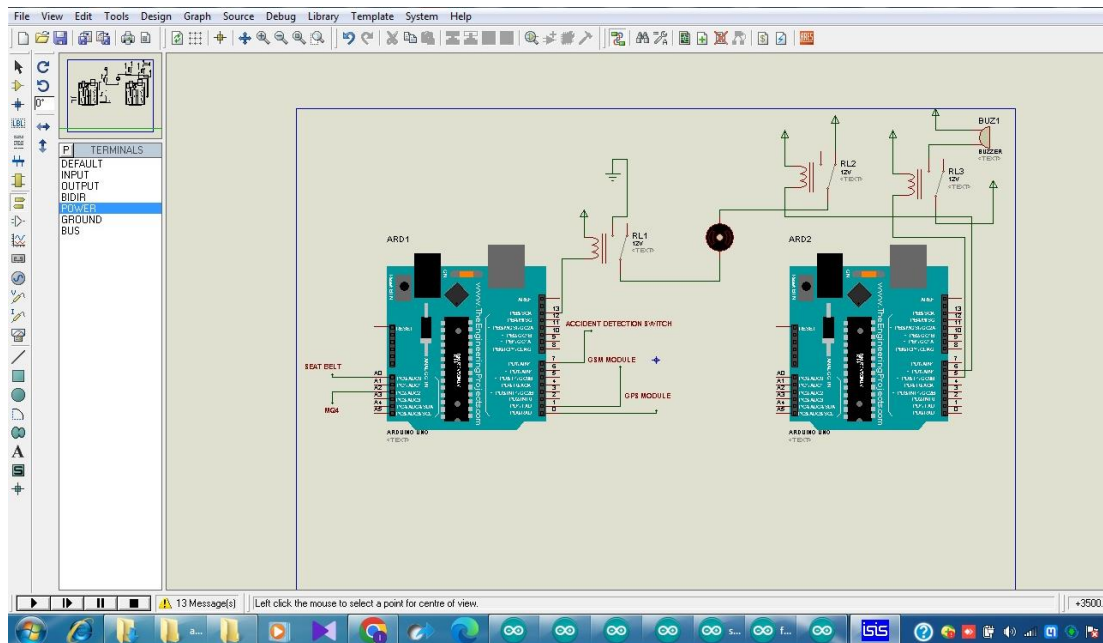


Figure 5
Circuit Diagram

3.5 Programming

One of the most important steps in the Internet of Things-based smart vehicle security and safety system is programming the Arduino microcontroller board. In order to allow the Arduino to communicate with different sensors, gather data, and send it to the chosen cloud platform, C/C++ code must be written. This is accomplished by using the Arduino Integrated Development Environment (IDE), which offers a user-friendly interface for uploading and coding to the microcontroller. To successfully integrate and utilize sensor functionalities inside the Arduino environment, specialized libraries are required.

For instance, the MQ3 Gas Sensor library interfaces with alcohol gas sensors, the MPU6050 library handles the eyeblink and seatbelt sensors, the GPS Module library processes location of the vehicle, and the GSM Module library manages the notification message that will be send to the provided contact.

By providing pre-built functions and methods, these libraries simplify development and improve the productivity and efficiency of the programming stage.

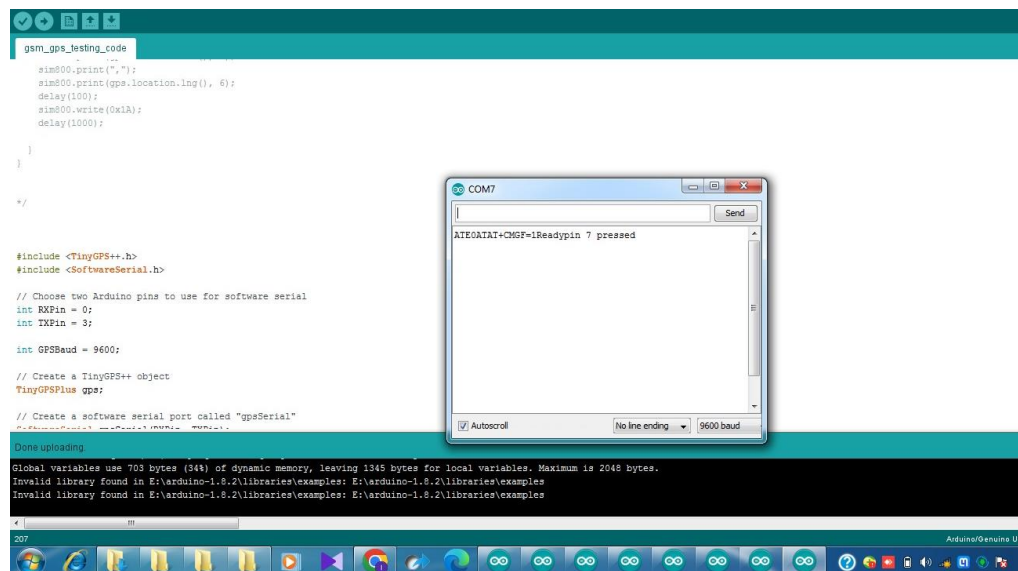


Figure 6

Arduino UNO Coding

3.6 Alert mechanism

We have included an alert mechanism in our Internet of Things-based smart vehicle security and safety system to let people know if there are any unusual circumstances or potential safety risks. The alert mechanism is composed of several kinds of components and features intended to provide prompt notification and response to possible threats.

Seat Belt Non-compliance Warning: The system sounds a warning if it notices that a passenger is not wearing their seat belt correctly while driving. By reminding passengers to buckle up, this alarm improves overall safety.

Drunk Driving Detection Alert: The alcohol sensor stops the engine from starting and sounds an alert if it detects that the driver has consumed alcohol. Ensuring the safety of all road users, this helps avoid incidents of drunk driving.

Drowsiness Detection Alert: The eye-blink sensor will sound an alert to remind the driver to take a break if it notices any indications of weariness or sleepiness. By taking this preventive action, road safety is increased and accidents brought on by tired drivers are reduced.

Emergency Notification Alert: In the event of an emergency, the system's emergency notification capability will automatically send notifications to pre-specified contacts or emergency services. This function ensures quick response in an emergency by using GPS technology to determine the location of the car.

Our system integrates various alarm mechanisms in an effort to improve road safety, avoid collisions, and improve emergency response times, all of which lead to better transportation conditions.

Chapter 4: Results and Discussion

We were able to obtain important insights into our IoT-based smart vehicle security and safety system's performance in a number of crucial sectors by means of its effective deployment and operation. Particularly, our experiment showed how effective it is at improving road safety, boosting vehicle security, and reducing the risks connected to reckless driving.

4.1 Detection of Seat Belt Non-Compliance

As our system effectively detected that when the car was moving, our technology successfully identified the instances of seat belt non-compliance and warned the occupants to buckle up. Our solution continuously tracks seat belt usage in real time by integrating sensors within the seat belt mechanism. The system sounds an alert to remind passengers to buckle up if it detects non-compliance. This feature increased adherence to seat belt laws, which lowers the possibility of injuries in the event of an accident and improves road safety widely.

4.2 Drunk Driving Avoidance

Our system's integration of an alcohol sensor was intended to address the major issue of drunk driving and ensure the security of both drivers and other road users. Here's how we managed to accomplish this goal. We added an alcohol sensor to the car's system, which is a piece of equipment that can identify alcohol in the breath of the driver or in the surrounding air. This sensor constantly monitors the air for the presence of alcohol vapours in real time. The system reacts when the sensor finds alcohol concentrations above a predetermined threshold. In order to stop an inebriated driver from operating the vehicle, this mechanism automatically disables the ignition. Our goal in being proactive was to lower the likelihood of driving while intoxicated accidents. In addition to improving safety, this policy may even save lives by preventing intoxicated drivers from operating a vehicle. This feature not only improves safety but also educates drivers about the risks associated with drunk driving.

4.3 Drowsiness Detection and Alert

Our system also includes an eye-blink sensor to help identify when drivers are becoming fatigued or drowsy. This sensor monitors the driver's blinking patterns to detect any indications of fatigue. The sensor isn't simply placed there; it reacts when it detects these kinds of signals. It immediately notifies the driver to take a break by sending out alerts. This prompt reaction is essential because it reduces the likelihood of accidents caused by fatigued drivers losing their concentration. Our approach also seeks to increase everyone's safety on the roadways by alerting drivers to take rests when necessary. A rested driver is, after all, far less likely to be involved in an accident. Also, by lowering the possibility of fatigued driving, we contribute to an increase in general traffic safety.

4.4 Collision Detection and Emergency Notification

As we install switches in our system that can detect collisions or accidents, we ensured that our system was prepared to react quickly in the event of an accident. These switches work like a red flag, alerting the system to react as soon as they detect an impact. It quickly notifies the individuals we've already designated to be contacted in the event of an emergency. Family members, close friends, or even emergency services like the police or ambulance might come under this classification. This prompt action is essential because it allows assistance to reach the place more quickly, possibly saving lives. After all, every moment matters during an emergency. Our approach intends to actually make a difference in people's safety when driving by getting help there as soon as possible. The result of the emergency notification is given as below;

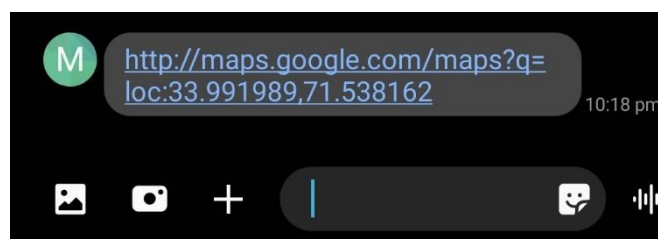


Figure 7

Emergency Notification Message

4.5 Alcohol/drugs, speed and fatigue in fatal crashes (2020 - 2022)

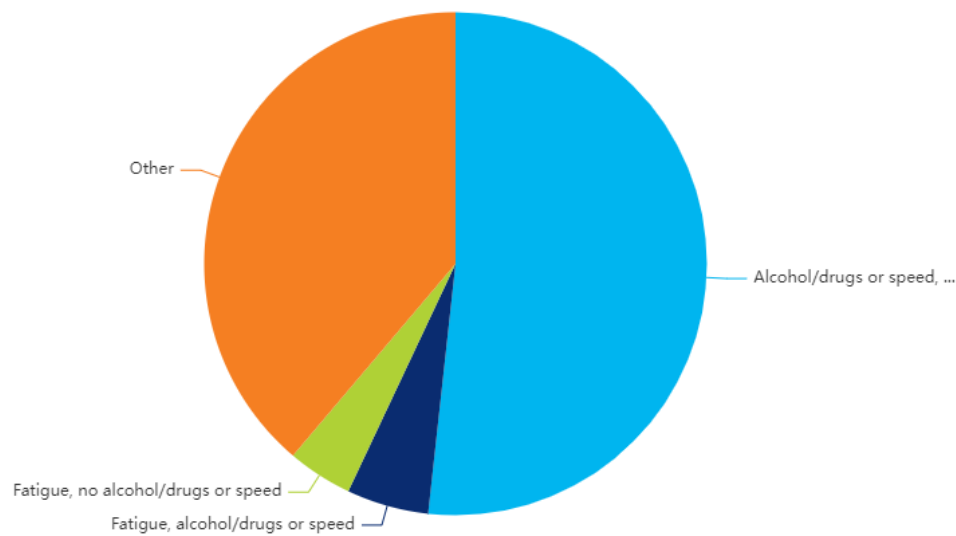
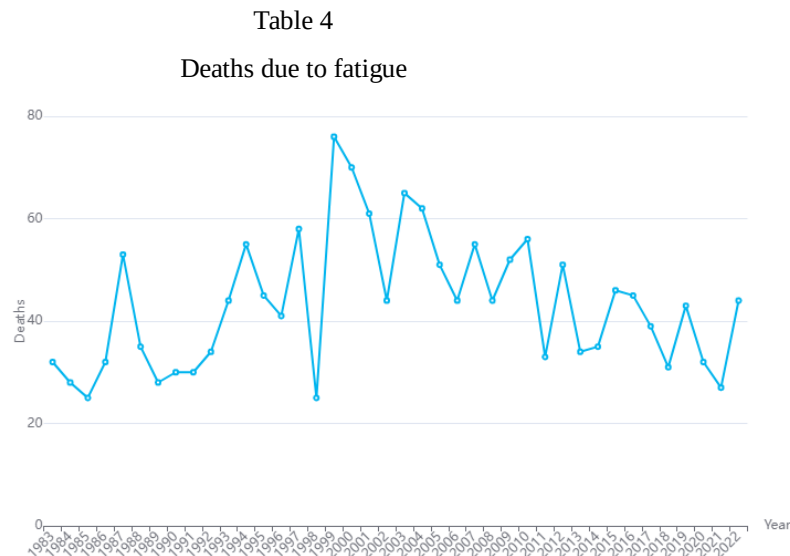


Figure 8

4.6 Deaths in crashes involving driver fatigue since 1983

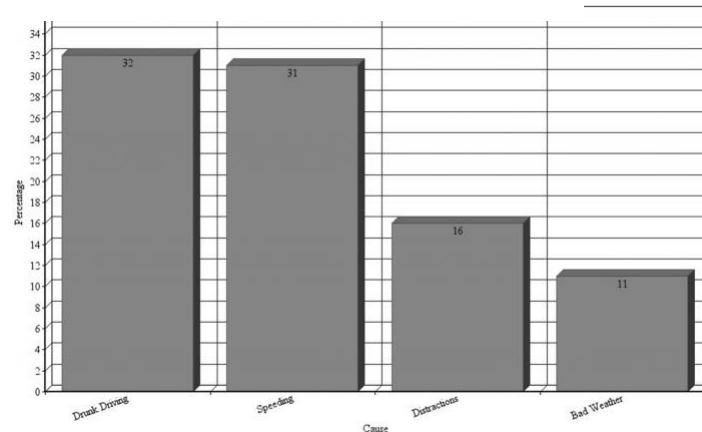


4.7 Causes of Crashes

The causes of accidents include drunk driving which is on the top of the list, the second is the over speeding that is also a cause of accident. The third cause of

accident was drowsiness of the driver and the fourth one was the bad weather conditions due to which there may have accident happens. The table of the main causes of accidents is given as below:

Table 5
Causes of Accidents



4.8 Discussion

4.8.1 Integration of Results with Project Objectives

Our IoT-based smart car safety and security system's outcomes closely match the project's goals. We were able to identify and stop seat belt violations, track down drunk drivers, keep an eye on driver fatigue, and deliver crash alerts and emergency warnings. This proves that our solution is successful in accomplishing its stated objectives of improving car security and encouraging traffic safety.

4.8.2 Limitations and challenges

Despite the encouraging outcomes of our IoT-based smart car safety and security system, there are a number of inherent limitations and challenges that must be taken into mind. These include the precision of the sensor, the dependability of the data transfer, the need for frequent maintenance and charging, the vulnerability to damage, and other possible problems.

Sensor Accuracy: One of the main issues is making sure the sensors that are integrated into our system are accurate and precise. Any inconsistencies in sensor readings or calibration mistakes might compromise the accuracy of the

information gathered and have an impact on the security and safety system's overall performance. To ensure that the system operates as intended and maintain high levels of accuracy, ongoing calibration and validation are required.

Data Transmission Reliability: It is essential that the sensors' data be reliably transmitted to the cloud platform or central monitoring unit. Data reception that is accurate and timely may be compromised by problems including transmission delays, network connectivity problems, or interference with signals. This is especially crucial for real-time warnings and monitoring, since any delay could have serious consequences.

Maintenance Requirements: Maintaining the monitoring system at its best requires routine maintenance. This covers maintenance activities like sensor adjustments, battery changes, software upgrades, and comprehensive system checks to quickly find and fix any possible problems. In the absence of routine maintenance, the system's accuracy and dependability could decrease over time, potentially creating security weaknesses.

Regular Charging: The system as a whole, as well as the sensors, depend on batteries that need to be charged frequently. The inability to keep adequate battery levels could result in data collecting disruptions and system outages, which would affect the system's capacity to deliver real-time security and safety insights. Maintaining a steady power source is essential for continuous functioning, particularly in emergency scenarios.

Overcoming these constraints and difficulties requires proactive strategies, continual observation, and continuous improvements. The long-term sustainability, usefulness, and dependability of our smart car security and safety system in real-world applications depend heavily on these initiatives. By resolving these issues, we can make sure that our system keeps offering insightful data and promotes general traffic safety and vehicle security.

Chapter 5: Conclusions and future work

5.1 Conclusion

To sum up, our project has created and executed an Internet of Things (IoT)-based smart vehicle security and safety system that achieves important goals including preventing drunk driving, detecting drowsiness, and encouraging seatbelt use. This system offers real-time insights into driver behaviour and vehicle conditions by utilizing cutting-edge sensor technology and IoT solutions. This allows for the implementation of preventive measures aimed at improving road safety. The outcomes show how well the system implements seatbelt usage, identify alcohol consumption, and keep an eye on driver attention. The system's usability, accessibility, and scalability are improved by the integration of continuous cloud storage and real-time data access via mobile applications, setting it apart from conventional car safety systems.

Furthermore, our approach also highlights emergency response and general road safety by alerting contacts in the event of an accident. Our system adds to a safer and more responsible driving environment by encouraging safe driving attitudes. While significant advancements have been made, it is important to address certain limitations and challenges, such as sensor accuracy, data transmission consistency, and system maintenance. Solving these challenges will be crucial for ensuring the continued reliability and efficiency of the system.

Looking forward, future research prospects are common and include developing machine learning algorithms, integrating with other vehicle safety systems, and advancing sensor technology. IoT-based vehicle safety systems will continue to be improved and creative as long as research in these fields continues to take place.

Overall, all things considered, this initiative is a major advancement in the field of vehicle safety, providing practical solutions to improve traffic safety and driver well-being. Through the application of technology, we hope to enable drivers with up-to-date knowledge and contribute to the global advancement of driving safety.

5.2 Summary of results

Throughout our project, we achieved significant milestones in the development and implementation of an IoT-based smart vehicle security and safety system. Here is a summary of the key results obtained:

5.2.1 Seat Belt Usage Compliance

By stopping vehicle starting until all seatbelts are buckled, our method efficiently assures compliance with seatbelt usage. In the event of an accident, this feature greatly lowers the chance of injury.

5.2.2 Drunk Driving Prevention

Using advanced sensors, the system precisely measures the driver's alcohol content and stops the car from starting if the driver is under the influence of alcohol. This feature contributes to a decrease in drunk driving accidents.

5.2.3 Drowsiness Detection

Eye-blink sensors are used to track drivers' levels of attention. Prompt detection of driver fatigue is made possible by real-time data analysis, which permits prompt action to avoid accidents.

5.2.4 Emergency Response

The technology automatically notifies pre-designated emergency contacts in the event of an accident, giving vital information for prompt medical care. This function improves the response time after an incident and could potentially save lives.

In general, the effectiveness of our monitoring system in providing instantaneous insights into the driving behaviour and situation of vehicles has been demonstrated. We are providing drivers with useful data to enhance safety and encourage responsible driving habits by utilizing IoT technologies.

5.3 Future Work and Research

Future work on the IoT based smart vehicle security and safety system will focus on enhancing real-time data processing through advanced algorithms and machine learning, integration additional environmental and biometric sensors

for comprehensive monitoring and improving communication protocols for low latency and high reliability.

5.3.1 Addressing Mortality in Road Accidents

There are still a lot of fatal collisions caused by careless and intoxicated drivers. Enhancing the IoT-based system to fully handle these problems and give drivers useful preventive measures should be the main focus of this research.

5.3.2 Sensor Optimization

More specifically, in difficult scenarios like high humidity and temperatures, extra effort is required to make sure that sensors are extremely precise and trustworthy. Investigating different kinds of sensors and ways to calibrate them will make it possible for the system to operate well in challenging circumstances.

5.3.3 Data Analytics and Predictive Modelling

It might be revolutionary to increase the system's intelligence through predictive modelling and sophisticated data analytics. The ability to forecast long-term driving trends, possible hazards, and the best safety precautions will be extremely helpful to drivers. System capabilities will be further improved by utilizing AI and machine learning to find patterns and connections in the data that has been gathered.

5.3.4 Integration with Advanced Vehicle Technologies

Safety and effectiveness can be increased even further by integrating the monitoring system with other advanced automotive technology like automatic emergency response systems, electronic vehicle identification, and autonomous driving capabilities. Collaborative systems have the potential to improve production, decrease wages, and improve overall vehicle operations.

FYDP to CEP Mapping

S#	Attribute	Complex Problem	Targeted (Yes/No)	Justification
1.	Preamble	*Engineering problems which cannot be resolved without in-depth engineering knowledge. And have some or all of the characteristics listed below:	Yes	This FYP idea involves using IoT technology to enhance vehicle safety and security, which requires in-depth engineering knowledge.
2.	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues.	Yes	This project likely involves conflicting requirements, such as ensuring vehicle safety while not compromising on the vehicle's performance or user experience.
3.	Depth of Analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.	Yes	This project likely involves conflicting requirements, such as ensuring vehicle safety while not compromising on the vehicle's performance or user experience.
4.	Depth of Knowledge required	Requires research-based knowledge much of which is at, or informed by, the forefront of the professional discipline and which allows a fundamental- based, first principles analytical approach.	Yes	Implementing an IoT-based system for vehicle safety and security demands a strong foundation in research-based knowledge, particularly in IoT technology and automotive safety systems.

5.	Familiarity of issues	Involve infrequently encountered issues	Yes	This project addresses issues related to road safety and IoT integration. While these issues may not be entirely new, the application of IoT to tackle them is less common and therefore involves infrequently encountered issues.
6.	Extent of application codes	Are outside problems encompassed by standards and codes of practice for professional engineering.	Yes	Since you are dealing with IoT integration for vehicle safety, your project may involve aspects that are not entirely covered by existing standards and codes of practice, making it outside the realm of conventional solutions.
7.	Extent of stakeholder involved	Involve diverse groups of stakeholders with widely varying needs.	Yes	This project involves multiple stakeholders, including vehicle owners, traffic authorities, and potentially emergency services.
8.	Consequences	Have significant consequences in a range of contexts.	Yes	Addressing vehicle safety and security through IoT has significant consequences in terms of potentially saving lives and reducing the severity of accidents.

9.	Interdependence	Are high level problems including many component parts or sub problems.	Yes	This project is likely to involve many component parts, such as sensors, communication systems, and data analysis algorithms, which are interconnected to create an effective vehicle safety system. This demonstrates interdependence.
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FYDP to SDGs Mapping

Sustainable Development Goals (SDGs)		
Goal#	Description	Justification
Goal 1	NO POVERTY End poverty in all its forms everywhere	N/A
Goal 2	ZERO HUNGER End hunger, achieve food security and improved nutrition and promote sustainable agriculture	N/A
Goal 3	GOOD HEALTH & WELL-BEING Ensure healthy lives and promote well-being for all at all ages	This project aims to enhance vehicle safety and security, which ultimately contributes to the well-being of individuals on the road. By using IoT technology to prevent accidents and respond to them effectively, your project aligns with the objective of ensuring healthy lives and promoting well-being by reducing the risks associated with road accidents and their consequences. Improved road safety can lead to a significant

		reduction in injuries and fatalities, directly contributing to Goal 3 of the SDGs.
Goal 4	QUALITY EDUCATION Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	N/A
Goal 5	GENDER EQUALITY Achieve gender equality and empower all women and girls	N/A
Goal 6	CLEAN WATER AND SANITATION Ensure availability and sustainable management of water and sanitation for all	N/A
Goal 7	AFFORDABLE AND CLEAN ENERGY Ensure access to affordable, reliable, sustainable, and modern energy for all	N/A
Goal 8	DECENT WORK AND ECONOMIC GROWTH Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all	N/A
Goal 9	INDUSTRY, INNOVATION, AND INFRASTRUCTURE Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.	N/A
Goal 10	REDUCED INEQUALITIES Reduce inequality within and among countries	N/A
Goal 11	SUSTAINABLE CITIES AND COMMUNITIES Make cities and human settlements inclusive, safe, resilient and sustainable	N/A
Goal 12	RESPONSIBLE CONSUMPTION AND PRODUCTION Ensure sustainable consumption and production patterns	N/A
Goal 13	CLIMATE ACTION Take urgent action to combat climate change and its impacts	N/A
Goal 14	LIFE BELOW WATER Conserve and sustainably use the oceans, sea and marine resources for sustainable development	N/A
Goal 15	LIFE ON LAND Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably	N/A

	manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss	
Goal 16	PEACE, JUSTICE AND STRONG INSTITUTIONS Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	N/A
Goal 17	PARTNERSHIPS Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development	N/A

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